

MATHEMATICS
Modern Algebra
Suggested exercises

- (1) (Jan 10) Exercises 1, 3, 5, 8, 9, 10 page 213.
- (2) (Jan 12) a) Let A be a commutative ring and M an A -module. M is Noetherian if any submodule of M is finitely generated or equivalently if it satisfies the ascending chain condition. Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be a short exact sequence. Show that M is Noetherian if and only if M', M'' are Noetherian.
 - b) Let $u : M \rightarrow M$ be a surjective homomorphism of A -modules. If M is Noetherian, show that u is an isomorphism. (Hint: consider $\text{Ker } u^m$ for all $m > 0$.) Give an example of a non Noetherian A -module M where this fails.
 - c) If A is a commutative ring and $f = a_n x^n + \dots + a_0$ is a zero divisor in $A[x]$, then there is an element $0 \neq a \in A$ such that $aa_n = aa_{n-1} = \dots = aa_0 = 0$. (Hint: Pick $g = b_0 + \dots + b_m x^m$ of smallest degree such that $fg = 0$. Then as $a_n b_m = 0$, one sees that $a_n g = 0$ (as it has smaller degree and $a_n g f = 0$). Show that $a_{n-r} g = 0$ for $0 \leq r \leq n$.)
 - d) Show that $x^2 + 3x + 2$ is irreducible in $\mathbb{Z}[[x]]$ but not in $\mathbb{Z}[x]$.
 - e) Let A be a commutative ring, $f = a_n x^n + \dots + a_1 x + a_0 \in A[x]$. Then f is a unit if and only if a_0 is a unit and $a_1, \dots, a_n$ are nilpotent (i.e. $a_i^{n_i} = 0$ some $n_i > 0$). (Hint: you will need to show by induction that $a_n^{r+1} b_{m-r} = 0$ for $0 \leq r \leq m$ where $g = b_0 + \dots + b_m x^m$ and $fg = 1$.)
- (3) (Jan 19) Due on Jan 26.
 - (a) Find the partial fraction expression of
$$\frac{x^5 - 4x^2 + 2}{x^4 - 1}.$$
 - (b) For what values of a does $x^3 + a$ have a root in common with $x^2 + bx + 1$?
 - (c) Which of the following is a complete local ring (explain in detail and give complete proofs and references).
 - (i) $\mathbb{Z}_p[[x]]$;
 - (ii) $k[x_1, \dots, x_n]_{(x_1, \dots, x_n)}$;
 - (iii) $k(x_1, \dots, x_n)$.
- (4) (Jan 26) Due on Feb 2. Exercises 1, 2, 3, 4, 5, 6 on page 253.
- (5) (Jan 31) Exercises 7, 8, 10, 11 on page 253.
- (6) (Feb 8) Exercises 9, 12, 14, 16, 19, 20 on page 253.

- (7) (Feb 14) Exercises 1 page 321 Due Feb 21, Ex. 2, 3 recommended.
- (8) (Feb 23) Due Feb 30.
 - (a) (Prelim Aug. 03) Suppose that F is a field whose multiplicative group is periodic. Prove that for some prime p , F is isomorphic to a subfield of the algebraic closure of the field of order p .
 - (b) (Prelim Aug. 03) Let G be the Galois group of the polynomial $x^3 - 3x^2 - 3$ over the field F . Determine G when $F = \mathbb{C}$, $F = \mathbb{R}$, $F = \mathbb{Q}$ and F is the field of order 2.
 - (c) (Prelim Aug. 03) Prove that there is a rational polynomial whose Galois group is the direct product of two groups of order 3.
 - (d) (Prelim Sept. 96) Let E be a finite extension of the rationals, Show that the number of roots of unity in E is finite.
 - (e) (Prelim Aug. 02) Determine for which fields the polynomial $X^3 - X + 1$ has a multiple root.
- (9) (Feb 23) Not due in class, but recommended: Prelims from Sept. 96 # 4, 9, Aug. 04 # 7, 9, Jan. 04 # 10, Aug. 01 # 3, Aug 00 # 2, 3, 4, 10
- (10) (Apr 11) Due Apr 20. Exercises 1, 2, 3, 4, 5, 8, 9, 10 on page 545.
- (11) (Apr 22) Suggested problems: Exercises 6, 9, 11, 13, 14, 19, 20, 21, 22, 23 on page 567; Exercises 3, 4 on page 596. From Prelims, # 3 Jan 2004, # 2 Aug. 2003, # 10 Aug. 2001, # 11 Aug. 1999, # 2 Aug 1998, # 2 Sept 1997, #1 Sept. 1995, # 2 Sept 1994